

Problem 1

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Problem 1 (a) Write as an exponential with base 5. 7^{3x} .

Explanation. $7 = 5^{\log_5(7)}$, so

$$7^{3x} = \left(5^{\log_5(7)}\right)^{3x} = 5^{3x \log_5(7)}$$

(b) Write in terms of the natural logarithm. $\log_3(4)$.

Explanation. The change of base formula says $\log_a(u) = \frac{\log_b(u)}{\log_b(a)}$, so

$$\log_3(4) = \frac{\ln(4)}{\ln(3)}$$

(c) Expand the following: $\log_{1/2} \left(\frac{6x^5(2 + \tan(x))^x}{\sqrt[5]{e^{4x} + 1}} \right)$.

Explanation.

$$\log_{1/2} \left(\frac{6x^5(2 + \tan(x))^x}{\sqrt[5]{e^{4x} + 1}} \right) = \log_{1/2}(6) + 5 \log_{1/2}(x) + x \log_{1/2}(2 + \tan(x)) - \frac{1}{5} \log_{1/2}(e^{4x} + 1)$$